

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EX-8403-GS

B.E. VIII Semester

Examination, December 2020

Grading System (GS)

Advanced Control System

(Elective - IV)

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Derive expression for the Euler-Lagrange's equation. Discuss the significance and applications of this equation.
2. a) Explain controllability and observability?
b) Check the controllability of a system having following coefficient matrices.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

3. a) Explain state variable feedback as applied to pole placement.
b) Explain the concepts of state variables.
4. Write a short note on salient feature of variable structure control. Explain with examples.

EX-8403-GS

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[2]

5. a) State and explain the Lyapunov stability theorem.
b) Consider a non-linear system described by the equations

$$\dot{x}_1 = x_2;$$

$$\dot{x}_2 = -(1 - |x_1|)x_2 - x_1;$$

find the region in the state-plane for which the equilibrium state of the system is asymptotically stable.

6. a) Distinguish between conventional control theory and modern control theory.
b) Show that the system described below is unstable

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 3 & -2 \end{bmatrix} x$$

verify the result using Lyapunov method.

7. Show that the scalar function $V(x)$ given below is positive definite and verify $-V(x)$ for negative definiteness.

$$V(x) = 8x_1^2 + 5x_2^2 + x_3^2 + 4x_1x_2 - 2x_1x_3 - 3x_2x_3$$

8. Write short notes on :
a) Pontryagin's maximum principle
b) Phase plane technique

EX-8403-GS